

PRODUCT : **TFT TOUCH MODULE****MODULE NO. :** **WKS50WV056-WCT****SUPPLIER:** **WKS Technology Co.,LTD****DATE:** **Jan 5, 2024**

SPECIFICATION

*Revision: 0.0***WKS50WV056-WCT***This module uses ROHS material*

This specification may change without prior notice in order to improve performance or quality. Please contact WKS R&D department for updated specification and product status before design for this product or release of this order.

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REVISION RECORD

REV NO.	REV DATE	CONTENTS	REMARKS
0.0	2024-01-05	First release	Preliminary

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1、GENERAL INFORMATION

Item of general information	Contents		Unit
LCD Display Size (Diagonal)	5.0		inch
Module Structure	LCD Display + CTP Touch + PCB		-
LCD Display Type	TFT		-
LCD Display Mode	Normally White		-
Recommended Viewing Direction	6		O'clock
Module size (W×H×T)	129.28×75.8×7.78		mm
Active area (W×H)	95.04×53.86		mm
Number of pixels (Resolution)	800RGB×480		pixel
Pixel pitch (W×H)	0.135 ×0.135		mm
Color Pixel Arrangement	RGB Stripe		-
LCD Driver IC	-		-
Module Interface Type	LCD	16bit/24bit RGB interface	-
	CTP	I2C interface	-
Module Input voltage	5.0V		V
Module Power Consumption	-		mW
Color Numbers	16.7M		-
Backlight Type	White LED		-

3、ABSOLUTE MAXIMUM RATINGS

Parameter of absolute maximum ratings	Symbol	Min	Max	Unit
Operating temperature	T_{op}	-20	70	℃
Storage temperature	T_{st}	-30	80	℃
Humidity	RH	-	90%(Max 60 ℃)	RH

Note: Absolute maximum ratings means the product can withstand short-term, not more than 120 hours. If the product is a long time to withstand these conditions, the life time would be shorter.

4、ELECTRICAL CHARACTERISTICS(DC CHARACTERISTICS)

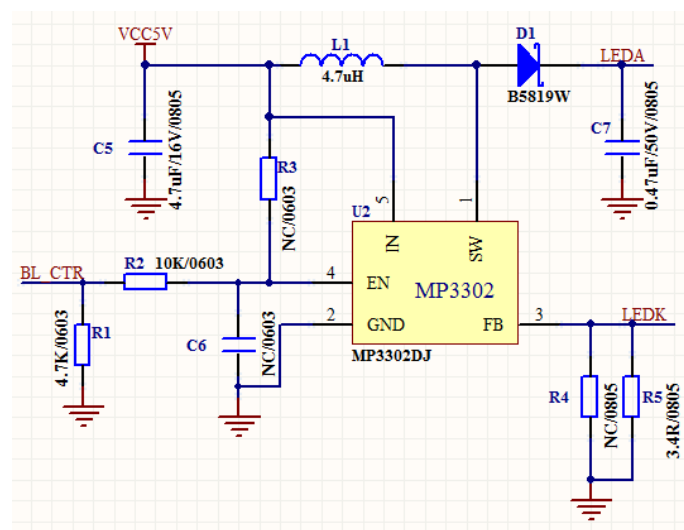
Parameter of DC characteristics	Symbol	Min.	Typ.	Max.	Unit
PCB operating voltage	V_{CC5V}	-	5.0	-	V
LCD I/O operating voltage	V_{DD}	2.7	3.3	3.6	V
Input voltage 'H' level	V_{IH}	$0.7*V_{DD}$	-	V_{DD}	V
Input voltage 'L' level	V_{IL}	V_{SS}	-	$0.3*V_{DD}$	V
Output voltage 'H' level	V_{OH}	$V_{DD}-0.4$	-	V_{DD}	V
Output voltage 'L' level	V_{OL}	V_{SS}	-	$V_{SS}+0.4$	V

5、BACKLIGHT CHARACTERISTICS

Item of backlight characteristics	Symbol	Min.	Typ.	Max.	Unit	Remark
Forward Voltage	V_f	4.5	5	5.5	V	Note1
Forward Current	I_f	-	60	-	mA	-
Number of LED	-	-	$6 \times 3 = 18$	-	Piece	-
LED Connection mode	S/P	-	Serial/Parallel	-	-	-
Lifetime of LED	-	-	10000	-	hour	Note2

Note:

- Note1: The LED Supply Voltage is defined by the number of LED at $T_a = 25^\circ\text{C}$ and $I_f = 60\text{mA}$.
- Note2: The LED lifetime define as the estimated time to 50% degradation of initial luminous. The LED lifetime could be decreased if operating I_f is larger than 40mA .
- Backlight control via the BL_CTR pin or PWM signal.
- Backlight circuit:



6、CTP CHARACTERISTICS

<i>Item of CTP characteristics</i>	<i>Specification</i>	<i>Unit</i>	<i>Remark</i>
<i>Panel Type</i>	<i>Glass Cover + Glass Sensor</i>	-	-
<i>Resolution</i>	<i>800 × 480</i>	<i>pixel</i>	-
<i>Surface Hardness</i>	<i>6H</i>	-	-
<i>Transparency</i>	<i>≥82%</i>	-	-
<i>Driver IC</i>	<i>GT911</i>	-	-
<i>Interface Type</i>	<i>I2C</i>	-	-
<i>Support Points</i>	<i>5</i>	-	-
<i>Sampling Rate</i>	<i>20~100</i>	<i>Hz</i>	-
<i>Supply voltage</i>	<i>3.3</i>	<i>V</i>	-

7、ELECTRO-OPTICAL CHARACTERISTICS

Item of electro-optical characteristics		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark	Note
Response time		Tr+Tf	$\theta=0$ $\phi=0$ $Ta=25\text{ }^{\circ}\text{C}$	-	25	50	ms	FIG 1.	4
Contrast Ratio		CR		-	500	-	-	FIG 2.	1
Luminance uniformity		$\delta WHITE$		-	80	-	%	FIG 2.	3
Surface Luminance		Lv		-	400	-	cd/m2	FIG 2.	2
CIE (x, y) chromaticity	White	White x	$\theta=0$ $\phi=0$ $Ta=25\text{ }^{\circ}\text{C}$	0.309	0.313	0.315	-	FIG 2.	5
		White y		0.337	0.339	0.341			
	Red	Red x		0.629	0.631	0.633			
		Red y		0.327	0.329	0.331			
	Green	Green x		0.326	0.328	0.330			
		Green y		0.546	0.548	0.550			
	Blue	Blue x		0.134	0.136	0.138			
		Blue y		0.139	0.141	0.143			
Viewing angle range	$\phi=90(12\text{ o'clock})$		CR ≥ 10	-	50	-	deg	FIG 3.	6
	$\phi=270(6\text{ o'clock})$			-	70	-	deg		
	$\phi=0(3\text{ o'clock})$			-	70	-	deg		
	$\phi=180(9\text{ o'clock})$			-	70	-	deg		
NTSC ratio		-	-	-	50	-	%	-	-

Note 1. Contrast Ratio(CR) is defined mathematically by the following formula. For more information see FIG 2.:

$$\text{Contrast Ratio(CR)} = \frac{\text{Average Surface Luminance with all white pixels(P1,P2,P3,P4,P5,P6,P7,P8,P9)}}{\text{Average Surface Luminance with all black pixels(P1,P2,P3,P4,P5,P6,P7,P8,P9)}}$$

Note 2. Surface luminance is the LCD surface from the surface with all pixels displaying white. For more information see FIG 2.

L_v =Average Surface Luminance with all white pixels (P1,P2,P3,P4,P5,P6,P7,P8,P9)

Note 3. The uniformity in surface luminance ($\delta WHITE$) is determined by measuring

luminance at each test position 1 through 9, and then dividing the maximum luminance of 9 points luminance by minimum luminance of 9 points luminance. For more information see FIG 2.

$$\delta_{\text{WHITE}} = \frac{\text{Minimum Surface Luminance with all white pixels (P1, P2, P3, P4, P5, P6, P7, P8, P9)}}{\text{Maximum Surface Luminance with all white pixels (P1, P2, P3, P4, P5, P6, P7, P8, P9)}}$$

Note 4. Response time is the time required for the display to transition from White to black(Rise Time, Tr) and from black to white(Decay Time, Tf). For additional information see FIG 1.

Note 5. CIE (x, y) chromaticity ,The x,y value is determined by screen active area position 5. For more information see FIG 2.

Note 6. Viewing angle is the angle at which the contrast ratio is greater than a specific value. For TFT module, the specific value of contrast ratio is 10.The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface. For more information see FIG 3.

Note 7. For Viewing angle and response time testing, the testing data is base on Autronic-Melchers's ConoScope. Series Instruments. For contrast ratio, Surface Luminance, Luminance uniformity and CIE, the testing data is base on BM-7 photo detector.

Note 8. For TN type TFT transmissive module, Gray scale reverse occurs in the direction of panel viewing angle.

FIG.1. The definition of Response Time

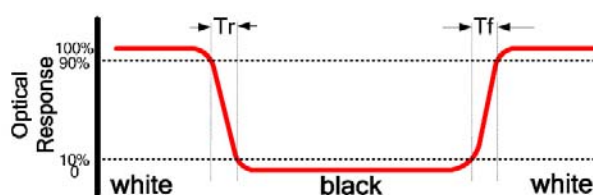


FIG.2. Measuring method for Contrast ratio, surface luminance, Luminance uniformity,

CIE (x , y) chromaticity

$A : H/6 ;$

$B : V/6 ;$

$H, V : \text{Active Area(AA) size}$

Measurement instrument: BM-7; Light spot size=5mm, 350mm distance from the LCD surface to detector lens.

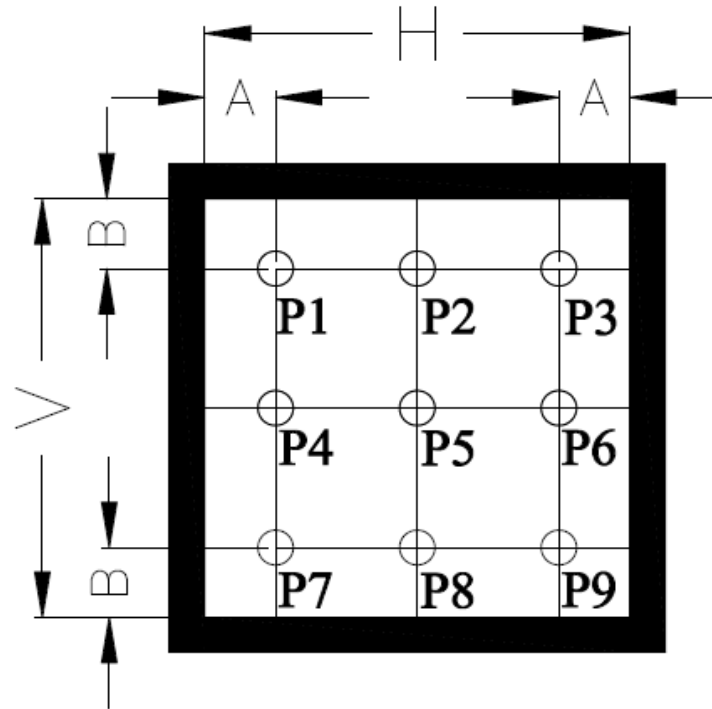
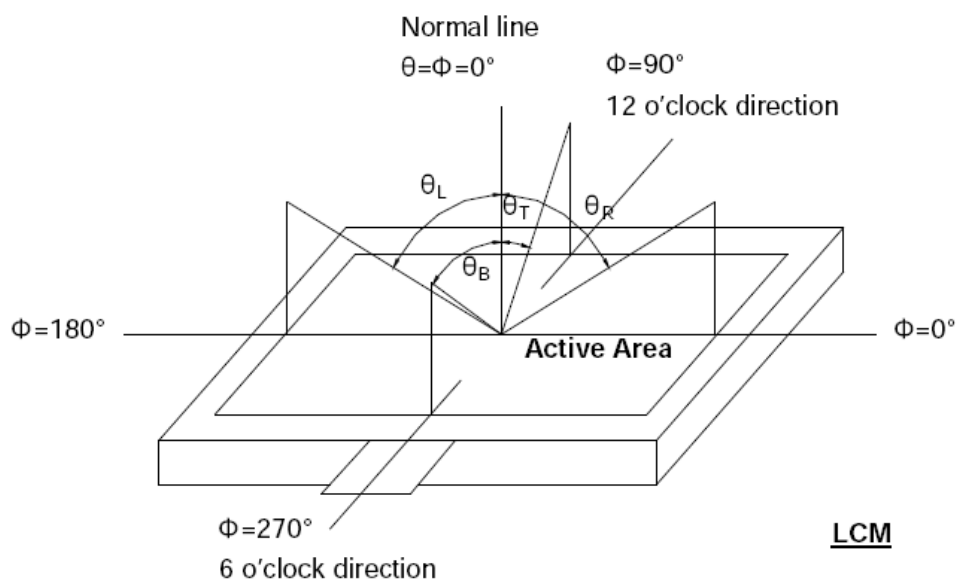


FIG.3. The definition of viewing angle



8、INTERFACE DESCRIPTION

8.1 Interface Description(LCD1)

NO.	Symbol	I/O	DESCRIPTION
1~2	VCC5V	Power supply	Module Power supply (5V Typ.)
3~10	R0~R7	I	8bit digital Red data input(R0:LSB; R7:MSB)
11	GND	Power supply	Power ground
12~19	G0~G7	I	8bit digital Green data input(G0:LSB; G7:MSB)
20	GND	Power supply	Power ground
21~28	B0~B7	I	8bit digital Blue data input(B0:LSB; B7:MSB)
29	GND	Power supply	Power ground
30	DCLK	I	Clock signal.
31	HSYNC	I	Horizontal Sync input.
32	VSNC	I	Vertical Sync input.
33	DEN	I	Data input Enable.
34	BL_CTRL	I	Backlight control pin
35	CTP_RST	I	CTP external reset signal, Low is active
36	CTP_SDA	I/O	CTP I2C data input and output
37	NC	-	No connection
38	CTP_SCL	I	CTP I2C clock input
39	CTP_INT	I	CTP External interrupt to the host
40	LCD_RST	-	No connection

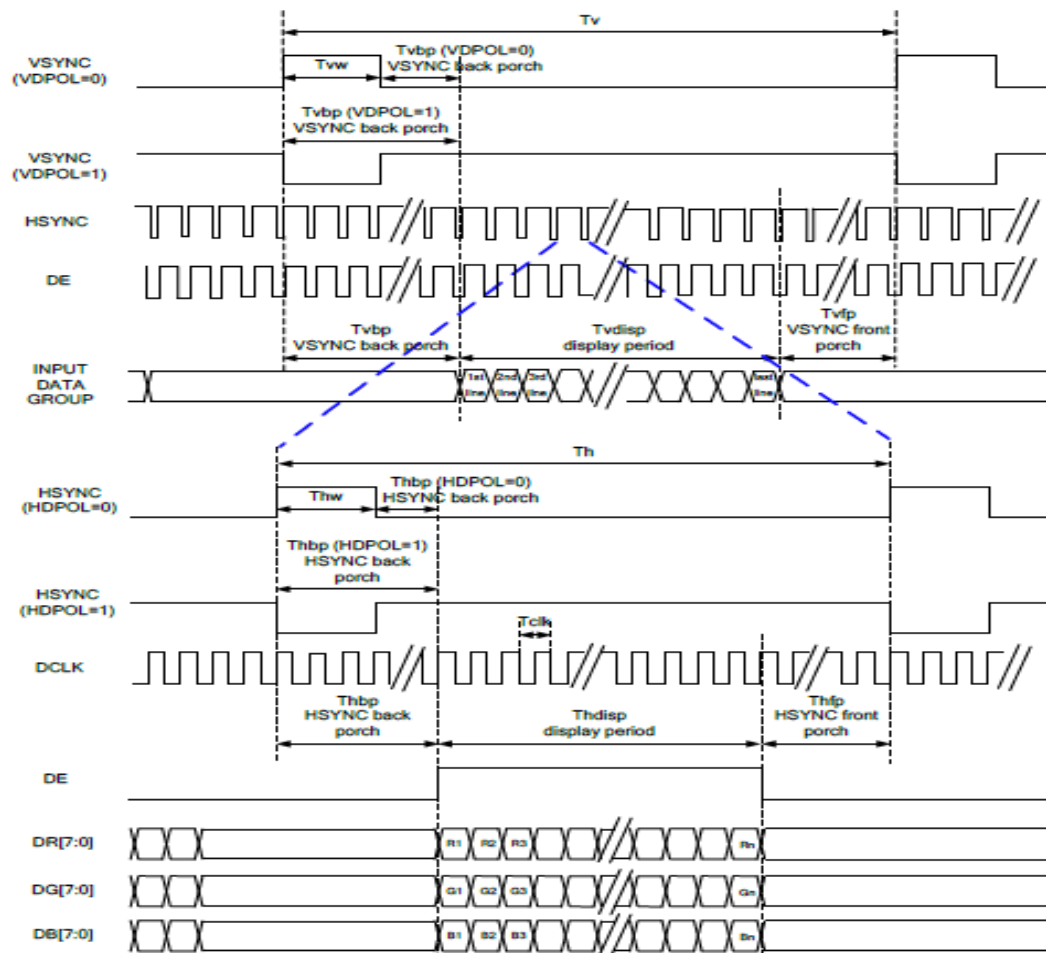
8.2 Interface Description(LCD2)

NO.	Symbol	I/O	DESCRIPTION
1~2	VCC5V	Power supply	Module Power supply (5V Typ.)
3~4	NC	-	No connection
5	GND	Power supply	Power ground
6~13	R0~R7	I	8bit digital Red data input(R0:LSB; R7:MSB)
14~21	G0~G7	I	8bit digital Green data input(G0:LSB; G7:MSB)
22~29	B0~B7	I	8bit digital Blue data input(B0:LSB; B7:MSB)
30	DCLK	I	Clock signal.
31	HSYNC	I	Horizontal Sync input.
32	VSYNC	I	Vertical Sync input.
33	DEN	I	Data input Enable.
34	NC	-	No connection
35	BL_CTRL	I	Backlight control pin
36	GND	Power supply	Power ground
37	CTP_RST	I	CTP external reset signal, Low is active
38	CTP_INT	I	CTP External interrupt to the host
39	CTP_SDA	I/O	CTP I2C data input and output
40	CTP_SCL	I	CTP I2C clock input

8.3 Interface Description(LCD3)

NO.	Symbol	I/O	DESCRIPTION
1	NC	-	No connection
2	VCC5V	Power supply	Module Power supply (5V Typ.)
3	GND	Power supply	Power ground
4~11	R0~R7	I	8bit digital Red data input(R0:LSB; R7:MSB)
12~19	G0~G7	I	8bit digital Green data input(G0:LSB; G7:MSB)
20~27	B0~B7	I	8bit digital Blue data input(B0:LSB; B7:MSB)
28	DCLK	I	Clock signal.
29	HSYNC	I	Horizontal Sync input.
30	VSYNC	I	Vertical Sync input.
31	DEN	I	Data input Enable.
32	NC	-	No connection
33	BL_CTRL	I	Backlight control pin
34	NC	-	No connection
35	CTP_INT	I	CTP External interrupt to the host
36	CTP_RST	I	CTP external reset signal, Low is active
37	CTP_SDA	I/O	CTP I2C data input and output
38	CTP_SCL	I	CTP I2C clock input
39~40	GND	Power supply	Power ground

9、INPUT TIMING



Parallel RGB input Timing table

Parameter	Symbol	Value			Unit
		Min.	Typ.	Max.	
PCLK frequency@ Frame rate=60Hz	PCLK	-	33.3	50	MHz
Horizontal display area	thd	800			PCLK
Horizontal period time	th	808	816	896	PCLK
HSYNC Back Porch	thbp	4	46	46	PCLK
HSYNC Front Porch	thfp	4	20	48	PCLK
HSYNC Pulse Width	thw	2	4	8	PCLK
Vertical display area	tvd	480			H
VSYNC period time	tv	488	496	504	H
VSYNC Back Porch	tvbp	4	10	12	H
VSYNC Front Porch	tvfp	4	20	147	H
VSYNC Pulse Width	tvw	2	4	8	H

10、RELIABILITY TEST CONDITIONS

No.	Test Item	Test Condition
1	High Temperature Storage	80 °C/120 hours
2	Low Temperature Storage	-30 °C/120 hours
3	High Temperature Operating	70 °C/120 hours
4	Low Temperature Operating	-20 °C/120 hours
5	Temperature Cycle Storage	-20 °C(30min.)~25(5min.)~70 °C(30min.)×10cycles

A、Inspection after test:

Inspection after 2~4 hours storage at room temperature, the sample shall be free from defects:

- Air bubble in the LCD;
- Sealleak;
- Non-display;
- Missing segments;
- Glass crack;
- Current is twice higher than initial value.

B、Remark:

- The test samples should be applied to only one test item.
- Sample size for each test item is 5~10pcs.
- Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

11、INSPECTION CRITERION

This specification is made to be used as the standard of acceptance/rejection criteria for TFT-LCD/IPS TFT-LCD module product, and this specification is applicable only in the case that the size of module equal to or exceed than 3.5 inch.

11.1 Sample plan

Sampling plan according to GB/T2828.1-2003/ISO 2859-1: 1999 and ANSI/ASQC Z1.4-1993,normal level 2 and based on:

Major defect: AQL 0.65

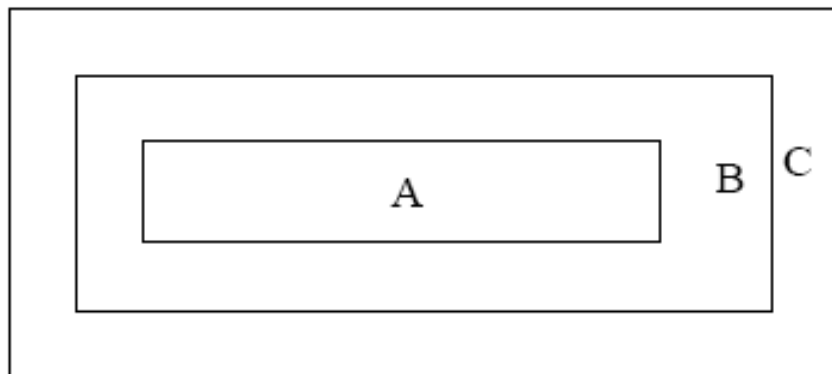
Minor defect: AQL 1.5

11.2 Inspection condition

Viewing distance for cosmetic inspection is about 30cm with bare eyes, and under an environment of 20~40W light intensity, all directions for inspecting the sample should be within 45° against perpendicular line. (Normal temperature 20~25 °C and normal humidity 60 ±15%RH)

11.3 Definition of Inspection Item

A、 *Definition of inspection zone in LCD.*



Zone A: character/Digit area

Zone B: viewing area except Zone A (Zone A + Zone B=minimum Viewing area)

Zone C: Outside viewing area (invisible area after assembly in customer's product)

Fig.1 Inspection zones in an LCD

Note: As a general rule, visual defects in Zone C are permissible, when it is no trouble for quality and assembly of customer's product.

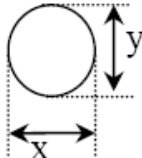
B、 Definition of some visual defect

<i>Bright dot</i>	<i>Because of losing all or part function, bad pixel dots appear bright and the size is more than 50% of one dot in which LCD panel is displaying under black pattern.</i>
<i>Dark dot</i>	<i>Dots appear dark and unchanged in size in which LCD panel is displaying under pure red, green, blue picture, or pure whiter picture.</i>

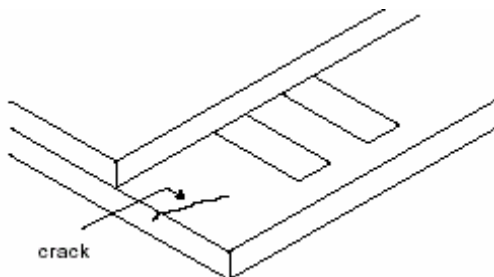
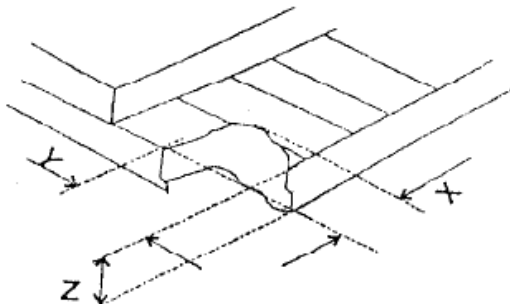
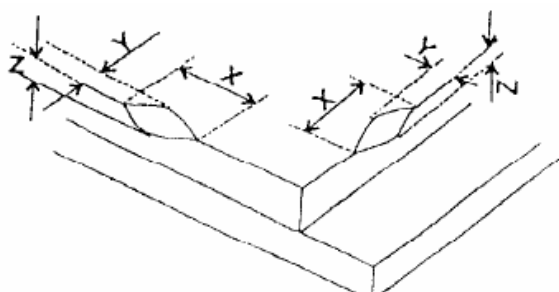
11.4 Major Defect

Item No.	Items to be inspected	Inspection standard	Classification of defects
<i>1</i>	<i>Functional defects</i>	<i>1) No display</i> <i>2) Display abnormally</i> <i>3) Missing vertical, horizontal segment</i> <i>4) Short circuit</i> <i>5) Excess power consumption</i> <i>6) Backlight no lighting, flickering and abnormal lighting</i>	<i>major</i>
<i>2</i>	<i>Missing</i>	<i>Missing component</i>	
<i>3</i>	<i>Outline dimension</i>	<i>Overall outline dimension beyond the drawing is not allowed</i>	

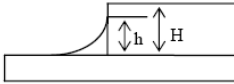
11.5 Minor Defect

Item No.	Items to be inspected	Inspection standard					Classification of defects	
1	Bright dot /dark dot defect	Zone		Acceptable Qty			Minor	
				A+B				
				3.5'' ~ 7''	7~10.1''	>10.1''		
		Bright pixel dot		1	2	3		Acceptable
		Dark pixel dot		4	4	4		
		2bright dots adjacent		0	0	0		
		2dark dots adjacent		0	0	0		
		Total bright and dark dots		5	6	7		
Note: Minimum distance between defective dots is more than 5mm; Pixel dots' function is normal, but bright dots caused by foreign material and other reasons are judged by the dot defect of 5.2.								
2	Dot defect  Φ=(x+y) /2	Zone		Acceptable Qty			Minor	
				A+B				
				3.5''~7''	7~10.1''	>10.1''		
		Φ≤0.2		Acceptable	Acceptable	Acceptable		Acceptable
		0.2 < Φ≤0.5		4	5	6		
		Φ >0.5		0	0	0		
Note: 1. Minimum distance between defective dots is more than 5 mm; 2. The quantity of defect is zero in operating condition.								
3	Linear defect	Zone		Acceptable Qty			Minor	
				A+B				
				Length	Width	3.5''~7''		7~10.1''
		Ignore	W≤0.05	Acceptable	Acceptable	Acceptable		Acceptable
		L≤5.0	0.05 < W≤0.1	4	5	6		
		L >5.0	W >0.1	0	0	0		

4	Polarizer defect	5.4.1 Polarizer Position (i) Shifting in position should not exceed the glass outline dimension. (ii) Incomplete covering of the viewing area due to shifting is not allowed. 5.4.2 Dirt on polarizer Dirt which can be wiped easily should be acceptable. 5.4.3 Polarizer Dent & Air bubble	<table><tr><th colspan="2" rowspan="2">Zone Size(mm)</th><th colspan="3">Acceptable Qty</th><th rowspan="3">C</th></tr><tr><th colspan="3">A+B</th></tr><tr><th>3.5"~7"</th><th>7~10.1"</th><th>>10.1"</th></tr><tr><td colspan="2">$\Phi \leq 0.2$</td><td>Acceptable</td><td>Acceptable</td><td>Acceptable</td><td rowspan="3">Acceptable</td></tr><tr><td colspan="2">$0.2 < \Phi \leq 0.5$</td><td>4</td><td>5</td><td>6</td></tr><tr><td colspan="2">$\Phi > 0.5$</td><td>0</td><td>0</td><td>0</td></tr></table>	Zone Size(mm)		Acceptable Qty			C	A+B			3.5"~7"	7~10.1"	>10.1"	$\Phi \leq 0.2$		Acceptable	Acceptable	Acceptable	Acceptable	$0.2 < \Phi \leq 0.5$		4	5	6	$\Phi > 0.5$		0	0	0	Minor	
		Zone Size(mm)				Acceptable Qty				C																							
				A+B																													
		3.5"~7"	7~10.1"	>10.1"																													
		$\Phi \leq 0.2$		Acceptable	Acceptable	Acceptable	Acceptable																										
		$0.2 < \Phi \leq 0.5$		4	5	6																											
		$\Phi > 0.5$		0	0	0																											
		5.4.4 Polarizer scratch (i) If the polarizer scratch can be seen after cover assembling or in the operating condition, judge by the linear defect of 5.3. (ii) If the polarizer scratch can be seen only in non-operating condition or some special angle, judge by the following:																															
		<table><tr><th colspan="2" rowspan="2">Zone Size (mm)</th><th colspan="3">Acceptable Qty</th><th rowspan="3">C</th></tr><tr><th colspan="3">A+B</th></tr><tr><th>Length</th><th>Width</th><th>3.5"~7"</th><th>7~10.1"</th><th>>10.1"</th></tr><tr><td>Ignore</td><td>$W \leq 0.05$</td><td>Acceptable</td><td>Acceptable</td><td>Acceptable</td><td rowspan="3">Acceptable</td></tr><tr><td>$1.0 < L \leq 5.0$</td><td>$0.05 < W \leq 0.20$</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$L > 5.0$</td><td>$W > 0.2$</td><td>0</td><td>0</td><td>0</td></tr></table>	Zone Size (mm)		Acceptable Qty			C	A+B			Length	Width	3.5"~7"	7~10.1"	>10.1"	Ignore	$W \leq 0.05$	Acceptable	Acceptable	Acceptable	Acceptable	$1.0 < L \leq 5.0$	$0.05 < W \leq 0.20$	4	5	6	$L > 5.0$	$W > 0.2$	0	0		0
		Zone Size (mm)			Acceptable Qty				C																								
A+B																																	
Length	Width	3.5"~7"	7~10.1"	>10.1"																													
Ignore	$W \leq 0.05$	Acceptable	Acceptable	Acceptable	Acceptable																												
$1.0 < L \leq 5.0$	$0.05 < W \leq 0.20$	4	5	6																													
$L > 5.0$	$W > 0.2$	0	0	0																													
5	MURA	Using 3% ND filter, it's NG if it can be seen in R,G,B picture.			Minor																												
	White/Black dot (MURA)	Visible under: ND3%; $D \leq 0.15\text{mm}$, Acceptable; $0.15\text{mm} < D \leq 0.5\text{mm}$, $N \leq 4$; $D > 0.5\text{mm}$, Not allowable.																															

6	Glass defect	(i) Crack Cracks are not allowed. 	Minor								
		(ii) TFT chips on corner  <table><tr><th>X</th><th>Y</th><th>Z</th><th>Acceptable</th></tr><tr><td>≤ 3.0</td><td>≤ 3.0</td><td>Not more than the thickness of glass</td><td>$N \leq 3$</td></tr></table> Chips on the corner of terminal shall not be allowed to extend into the ITO pad or expose perimeter seal.	X	Y	Z	Acceptable	≤ 3.0	≤ 3.0	Not more than the thickness of glass	$N \leq 3$	Minor
		X	Y	Z	Acceptable						
≤ 3.0	≤ 3.0	Not more than the thickness of glass	$N \leq 3$								
(iii) Usual surface crack  <table><tr><th>X</th><th>Y</th><th>Z</th><th>Acceptable</th></tr><tr><td>≤ 1.5</td><td>≤ 1.5</td><td>Not more than the thickness of glass</td><td>$N \leq 4$</td></tr></table> It is only applicable to the upper glass of LCD.	X	Y	Z	Acceptable	≤ 1.5	≤ 1.5	Not more than the thickness of glass	$N \leq 4$	Minor		
X	Y	Z	Acceptable								
≤ 1.5	≤ 1.5	Not more than the thickness of glass	$N \leq 4$								

11.6 Module Cosmetic Criteria

Item No.	Items to be inspected	Inspection Standard	Classification of defects
1	Difference in Spec.	Not allowable	Major
2	Pattern peeling	No substrate pattern peeling and floating	Major
3	Soldering defects	No soldering missing	Major
		No soldering bridge	Major
		No cold soldering	Minor
4	Resist flaw on PCB	Visible copper foil ($\Phi 0.5$ mm or more) on substrate pattern is not allowed	Minor
5	FPC gold finger	No dirt, breaking, oxidation lead to black	Major
6	Backlight plastic frame	No deformation, crack, breaking, backlight positioning column breaking, obvious nick.	Minor
7	Marking printing effect	No dark marking, incomplete, deformation lead to unable to judge	Minor
8	Accretion of metallic Foreign matter	No accretion of metallic foreign matter (Not exceed $\Phi 0.2$ mm)	Minor
9	Stain	No stain to spoil cosmetic badly	Minor
10	Plate discoloring	No plate fading, rusting and discoloring	Minor
11	1. Lead parts	a. Soldering side of PCB Solder to form a 'Filet' all around the lead. Solder should not hide the lead form perfectly.	Minor
		b. Components side(In case of 'Through Hole PCB') Solder to reach the Components side of PCB.	Minor
	2. Flat packages	Either 'Toe'(A) or 'Seal'(B) of the lead to be covered by "Filet". Lead form to be assume over Solder. 	Minor
	3. Chips	$(3/2) H \geq h \geq (1/2) H$ 	Minor
	4. Solder ball/Solder splash	a. The spacing between solder ball and the conductor or solder pad $h \geq 0.13$ mm. The diameter of solder ball $d \leq 0.15$ mm.	Minor
		b. The quantity of solder balls or solder splashes isn't beyond 5 in 600 mm ² .	Minor
		c. Solder balls/Solder splashes do not violate minimum electrical clearance.	Major